

Damietta University Faculty of Computers and Artificial Intelligence	Mid Term Exam 2024-2025	15 Marks Course: Data Science Date: Wednesday
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Student name: Student ID:

Choose the Correct Answer (Answer THREE questions only)

- The most widely used programming languages by data scientists are
a) R, Python and SQL ✓ b) Java and JavaScript
c) C and C++ d) Ruby
- The original hypothesis is known as
a) Alternate hypothesis b) Null hypothesis ✓
c) Both a and b are correct d) Both a and b are incorrect
- All of the following are unstructured data EXCEPT
a) Emails b) Facebook posts
c) IOT outputs d) DB table ✓
- Which of the following techniques are used to handle noisy data?
a) Binning b) Regression
c) Clustering d) All of the mentioned ✓
- If two variables' values move in the opposite direction, the correlation is said to be
a) Non-linear b) Linear
c) Positive d) Negative ✓
- In bin means smoothing, each value of the bin is replaced by
a) Deviation of the values in the bin b) Mean of the values in the bin ✓
c) Mode of the values in the bin d) Median of the values in the bin
- Which of the following are data preprocessing?
a) Handling missing values b) Correcting inconsistent formats
c) Removing duplicates d) All the mentioned ✓
- Adjusting numeric values to a common scale in machine learning is called
a) Data normalization ✓ b) Data reduction
c) Data compression d) Data cleaning
- In smoothing by bin boundaries, each value in the bin is replaced by
a) Maximum value in the bin b) Mean value in the bin
c) Minimum value in the bin d) Closest value between the maximum and minimum value ✓

Question ONE: Answer the following

1. Discuss Data Pre-processing.

Data Pre-processing is the process of transforming raw data into a clean, consistent, and usable format before analysis or machine learning. It is a critical step because real-world data is often incomplete, noisy, or inconsistent.

Main steps of Data Pre-processing:

- **Data Cleaning:** Handle missing values (fill with mean/median or remove), remove duplicates, and fix inconsistencies.
- **Data Integration:** Combine data from multiple sources into a unified dataset.

- **Data Transformation:** Normalize or standardize data to bring features to the same scale.
- **Data Reduction:** Reduce the volume of data while keeping important information (e.g., PCA, feature selection).
- **Data Discretization:** Convert continuous data into categorical bins.
- **Feature Encoding:** Convert categorical variables into numerical format (e.g., one-hot encoding).

2. Give 3 examples of messy data, explain how to manipulate it?

Example 1 — Missing Values:

Some records have empty fields (e.g., a patient's age is missing in a medical dataset).
How to handle: Fill with mean/median/mode (imputation) or remove the row if not critical.

Example 2 — Duplicate Records:

The same entry appears more than once (e.g., a customer registered twice with slightly different spelling).
How to handle: Detect duplicates using unique keys and remove redundant rows.

Example 3 — Inconsistent Formats:

Data stored in different formats (e.g., dates written as '01/05/2024' and 'May 1, 2024' in the same column).
How to handle: Standardize all values to one consistent format during the cleaning step.

3. Complete the following table (Z-score Normalization):

Formula: $z = (x - \text{mean}) / \text{std}$ | Data: 3,8,11,15,17,20,30,130 mean = 29.25 std ≈ 38.85

data	z-score normalized data
3	$(3 - 29.25) / 38.85 \approx -0.676$
8	$(8 - 29.25) / 38.85 \approx -0.547$
11	$(11 - 29.25) / 38.85 \approx -0.470$
15	$(15 - 29.25) / 38.85 \approx -0.367$
17	$(17 - 29.25) / 38.85 \approx -0.315$
20	$(20 - 29.25) / 38.85 \approx -0.238$
30	$(30 - 29.25) / 38.85 \approx +0.019$
130	$(130 - 29.25) / 38.85 \approx +2.594$

Question TWO

1. The prices in thousands of dollars for 152 cars — Which histogram describes this data?

Price interval	[10,22)	[22,27)	[27,34)	[34,46)	[46,58)	[58,70)	[70,110)
No. of cars	26	26	30	29	14	14	13

The histogram that matches this data is histogram **(b)** — it shows a right-skewed distribution with the highest frequency around [27,34) and a long tail to the right, consistent with most cars priced in the lower range and few expensive ones.

2. Mention the six steps for data wrangling.

Discovery: Understanding the data, exploring sources, and identifying patterns or trends.

Structuring: Organizing data into a proper format (e.g., tables, columns).

Cleaning: Fixing errors like missing values, duplicates, and inconsistencies.

Enriching: Combining data from multiple sources to add more value and insights.

Validating: Checking data accuracy and ensuring it meets requirements.

Publishing: Saving or sharing the final prepared data for analysis or use.

3. Explain briefly p-value with example.

P-value: The probability of obtaining the observed results (or more extreme) assuming the null hypothesis H_0 is true.

- If $p\text{-value} < \text{significance level } (\alpha, \text{ usually } 0.05) \rightarrow \text{Reject } H_0$

- If $p\text{-value} \geq \alpha \rightarrow \text{Fail to reject } H_0$

Example: A drug company claims their pill reduces blood pressure. After a study, $p\text{-value} = 0.03$. Since $0.03 < 0.05$, we reject H_0 and conclude the drug has a significant effect.

4. Describe overfitting and underfitting in detail.

Overfitting: The model learns the training data too well — including its noise and random fluctuations — so it performs very well on training data but poorly on unseen test data. $\rightarrow$ Low bias, High variance

Underfitting: The model is too simple and fails to capture the underlying patterns in the data, performing poorly on both training and test data. $\rightarrow$ High bias, Low variance

Solution: Use cross-validation, regularization (Ridge/Lasso), more training data, or adjust model complexity.

5. Mention the difference between classification and regression?

Feature	Classification	Regression
Output	Discrete categories / labels	Continuous numerical values
Goal	Predict which class a sample belongs to	Predict a quantity or value
Example	Email: Spam or Not Spam	Predicting house price
Algorithms	Decision Tree, SVM, KNN	Linear Regression, SVR
Evaluation	Accuracy, Precision, Recall	MSE, RMSE, R^2 score

6. Given data: 18,22,6,6,9,14,20,21,12,18,18,16 Bin depth=3

Step 1 — Sort: 6, 6, 9, 12, 14, 16, 18, 18, 18, 20, 21, 22

Step 2 — Bins: Bin1: 6,6,9 | Bin2: 12,14,16 | Bin3: 18,18,18 | Bin4: 20,21,22

Bin	Values	Mean	Result (Mean Smoothing)
Bin 1	6, 6, 9	7	7, 7, 7
Bin 2	12, 14, 16	14	14, 14, 14
Bin 3	18, 18, 18	18	18, 18, 18
Bin 4	20, 21, 22	21	21, 21, 21

Bin	Min / Max	Result (Boundary Smoothing)
Bin 1	6 / 9	6, 6, 9

Bin 2	12 / 16	12, 12, 16
Bin 3	18 / 18	18, 18, 18
Bin 4	20 / 22	20, 20, 22

7. Explain null and alternative hypothesis by considering odd and even cases.

Null Hypothesis (H_0): The default assumption that there is no significant effect or difference. It represents the status quo.

Alternative Hypothesis (H_1): The claim we want to prove — that there IS a significant effect or difference.

Odd Case Example:

Claim: A coin is fair.

H_0 : $P(\text{Heads}) = 0.5$ | H_1 : $P(\text{Heads}) \neq 0.5$

If we flip 11 times and get 9 heads $\rightarrow$ p-value is small $\rightarrow$ reject H_0 .

Even Case Example:

Claim: A new drug reduces blood pressure.

H_0 : Drug has no effect | H_1 : Drug reduces blood pressure

If p-value $< 0.05 \rightarrow$ reject $H_0 \rightarrow$ drug is effective.

Question THREE

1. Confusion Matrix: TP=62, TN=57, precision=31/32, accuracy=119/130 — Find FP, FN, recall.

Step 1 — Find FP from Precision:

Precision = $TP / (TP + FP) = 31/32$

$62 / (62 + FP) = 31/32$

$62 \times 32 = 31 \times (62 + FP)$

$1984 = 1922 + 31 \cdot FP$

FP = $62/31 = 2$

Step 2 — Find FN from Accuracy:

Accuracy = $(TP + TN) / (TP + TN + FP + FN) = 119/130$

$(62 + 57) / (62 + 57 + 2 + FN) = 119/130$

$119 / (121 + FN) = 119/130$

FN = $130 - 121 = 9$

Step 3 — Calculate Recall:

Recall = $TP / (TP + FN) = 62 / (62 + 9) = 62/71 \approx 0.873$

TP	TN	FP	FN	Precision	Accuracy	Recall
62	57	2	9	$31/32 \approx 0.969$	$119/130 \approx 0.915$	$62/71 \approx 0.873$

2. Apply data discretization with ages:

1,5,9,4,7,11,14,17,13,18,19,31,33,36,42,44,46,70,74,78,77 $\rightarrow$ categories: child, young, mature.

Sorting: 1, 4, 5, 7, 9, 11, 13, 14, 17, 18, 19, 31, 33, 36, 42, 44, 46, 70, 74, 77, 78

Category	Age Range	Ages
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Child	0 – 12	1, 4, 5, 7, 9, 11
Young	13 – 35	13, 14, 17, 18, 19, 31, 33
Mature	36 and above	36, 42, 44, 46, 70, 74, 77, 78

3. Explain predictive modelling in detail.

Predictive modelling is a statistical/machine learning process that uses historical data to build a model that can predict future outcomes or unknown values.

Steps of Predictive Modelling:

1. **Define the problem:** Identify what needs to be predicted.
2. **Collect data:** Gather relevant historical data.
3. **Pre-process data:** Clean, transform, and prepare the data.
4. **Choose a model:** Select an appropriate algorithm (e.g., Linear Regression, Decision Tree).
5. **Train the model:** Fit the model on training data.
6. **Evaluate:** Test on unseen data using metrics like accuracy, RMSE, etc.
7. **Deploy:** Use the model to make real predictions.

Types: Regression (predict numbers) | Classification (predict categories) | Time-series (predict future values over time).

4. Skewness for data 2,4,6,6 using Pearson's first coefficient + Linear Regression.

Skewness:

Data: 2, 4, 6, 6

Mean = $(2+4+6+6)/4 = 18/4 = 4.5$

Mode = 6 (most frequent)

Std = $\sqrt{[(2-4.5)^2 + (4-4.5)^2 + (6-4.5)^2 + (6-4.5)^2]/4} = \sqrt{(6.25+0.25+2.25+2.25)/4} = \sqrt{(11/4)} \approx 1.658$

Pearson's 1st coefficient = $(\text{Mean} - \text{Mode}) / \text{Std} = (4.5 - 6) / 1.658 = -1.5/1.658 \approx -0.905$

Since skewness < 0 → **Negatively skewed (Left skewed)**

Simple Linear Regression:

Study Hours (x)	2	4	6	8	10
Study Score (y)	50	60	70	80	90

$n=5, \Sigma x=30, \Sigma y=350, x_{\text{mean}}=6, y_{\text{mean}}=70$

$\Sigma xy = (2 \times 50) + (4 \times 60) + (6 \times 70) + (8 \times 80) + (10 \times 90) = 100 + 240 + 420 + 640 + 900 = 2300$

$\Sigma x^2 = 4 + 16 + 36 + 64 + 100 = 220$

$b = (\Sigma xy - n \cdot x_{\text{mean}} \cdot y_{\text{mean}}) / (\Sigma x^2 - n \cdot x_{\text{mean}}^2) = (2300 - 5 \times 6 \times 70) / (220 - 5 \times 36) = (2300 - 2100) / (220 - 180) = 200/40 = 5$

$a = y_{\text{mean}} - b \cdot x_{\text{mean}} = 70 - 5 \times 6 = 70 - 30 = 40$

Equation: $y = 40 + 5x$

Predict for $x=5$: $y = 40 + 5 \times 5 = 40 + 25 = 65$

5. Write a Python program to compute median (odd and even).

```
def median(arr):
    arr = sorted(arr)
    n = len(arr)
    if n % 2 == 1:          # Odd case
        return arr[n//2]
    else:                  # Even case
```

```
        return (arr[n//2 - 1] + arr[n//2]) / 2
# Test
print(median([3, 1, 4, 1, 5]))    # Output: 3    (odd)
print(median([3, 1, 4, 2]))       # Output: 2.5  (even)
```

Best Wishes ♦